

ExCP: Checkpoint Compression

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~70×

Compression (Pythia-410M)

410M → 7B

Models Evaluated

Near-lossless

Training Recovery

Title

ExCP Framework Overview

Intro

Why compress LLM checkpoints?

Motivation

- LLM training saves frequent checkpoints; storage can reach TB–PB and huge cost.
- Adam moments double storage; prior works (LC-Checkpoint, DynaQuant) focus on weights.
- ExCP uses residuals, joint weight–momentum pruning, non-uniform quantization.
- Evaluated on 410M–7B; up to ~70× reduction with strong downstream performance.

Why Compress Checkpoints?

- Pythia-12B: >24 GB/checkpoint; 154 checkpoints $\approx$ 11 TB (~\$5k/month cloud storage).
- Adam moments $\approx$ 2 $\times$ weights $\rightarrow$ large overhead.
- Advanced LLMs (GPT, Gemini) with 100B–1T+ params exacerbate burden.

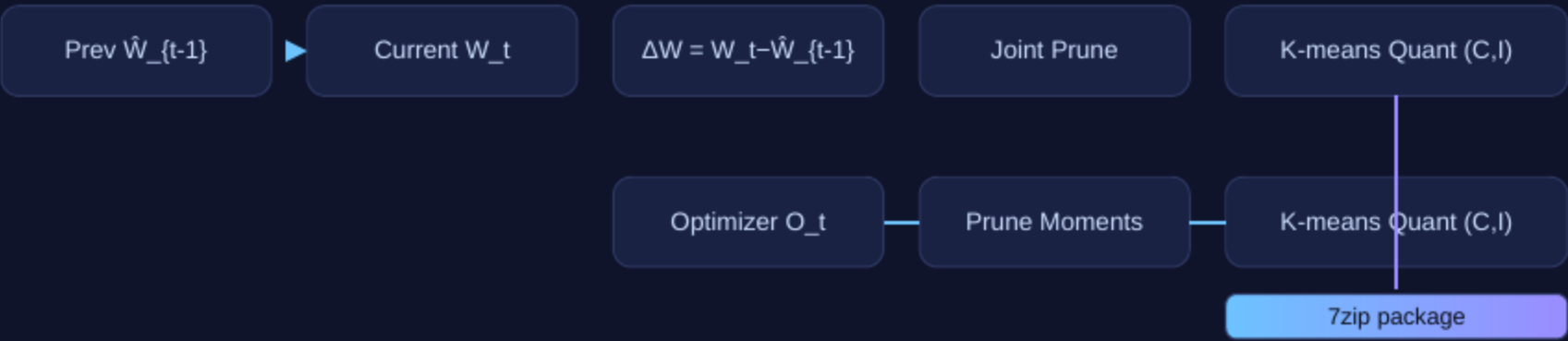
Figure 1: (a) LLM parameter scales; (b) Training stages and many checkpoints

Figure 1. LLM sizes and training process imply massive checkpoint storage.

Context

Scale and frequency drive storage

Framework



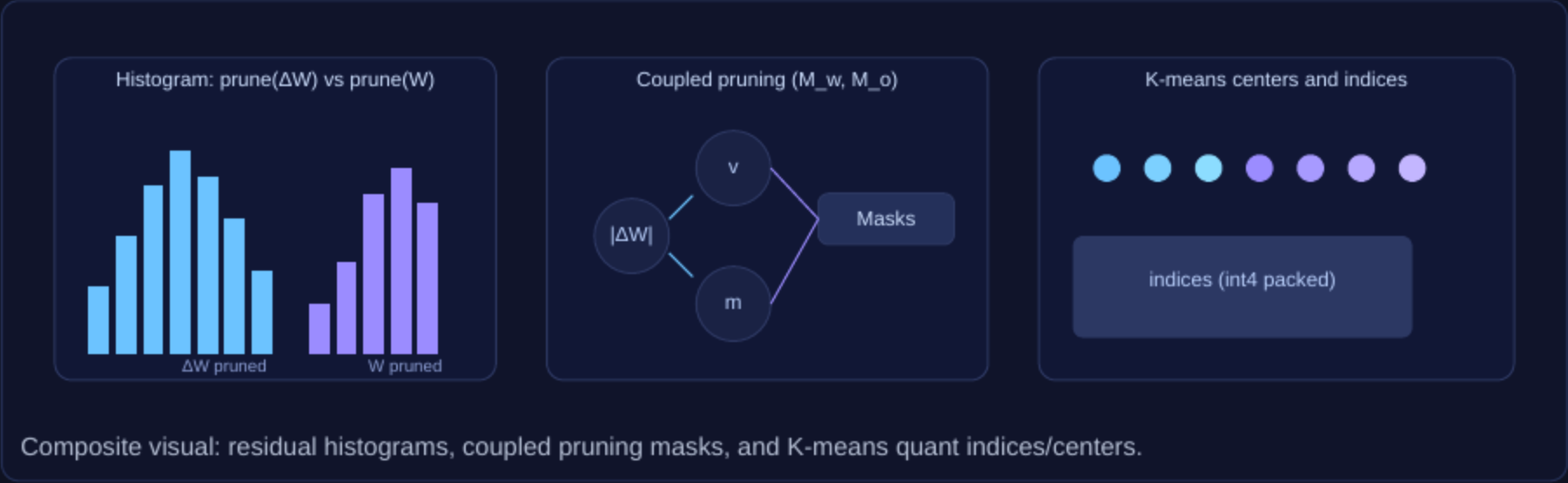
Pipeline: residual ΔW ; coupled pruning for W and moments; non-uniform K-means quantization; 7zip packaging.

Method

Residuals + Joint pruning + Quant

Method

- Residuals: store ΔW ; do not delta O (weak temporal link).
- Coupled pruning: thresholds guided by v (for W) and m (for O); share masks where W pruned.
- Convergence: Adam-style regret with small extra term from pruning small v ; empirical match.
- Quantization: K-means to 2^n-1 centers (zero kept); $n=4$ typical; pack and 7zip.



Method

Concise composite visual

Algorithms

Algorithm 1 Compress

```
Input: last reconstructed  $\hat{W}_{t-1}$ , original  $W_t$ , optimizer  $O_t$ 
 $\Delta W_t \leftarrow W_t - \hat{W}_{t-1}$ 
 $\Delta W^*_t, O^*_t \leftarrow \text{joint\_prune}(\Delta W_t, O_t)$ 
 $I^{\wedge}W_t, C^{\wedge}W_t \leftarrow \text{quantize}(\Delta W^*_t)$ 
 $I^{\wedge}O_t, C^{\wedge}O_t \leftarrow \text{quantize}(O^*_t)$ 
 $P'_t \leftarrow 7\text{zip}(I^{\wedge}W_t, C^{\wedge}W_t, I^{\wedge}O_t, C^{\wedge}O_t)$ 
save  $P'_t$ 
```

Algorithm 2 Reconstruct

```
Input: last reconstructed  $\hat{W}_{t-1}$ , compressed  $P_t$ 
 $I^{\wedge}W_t, C^{\wedge}W_t, I^{\wedge}O_t, C^{\wedge}O_t \leftarrow \text{unzip}(P_t)$ 
 $\Delta \hat{W}^*_t \leftarrow C^{\wedge}W_t[I^{\wedge}W_t]$ 
 $\hat{O}_t \leftarrow C^{\wedge}O_t[I^{\wedge}O_t]$ 
 $\hat{W}_t \leftarrow (\hat{W}_{t-1} \text{ or } W_{t-1}) + \Delta \hat{W}^*_t$ 
```

Algorithm 3 Reconstruct arbitrary

```
Input: seed  $s$ ,  $\{P_i\}$ , target  $t$ 
 $\hat{W}_0 = \text{init}(s)$ 
for  $i=1..t$ :  $\hat{W}_i \leftarrow \text{recon}(\hat{W}_{i-1}, P_i)$ 
```

Procedure

Independent compression; fast resume

Setup

- Models: ViT-L/32; Pythia-410M; PanGu π -1B; PanGu π -7B.
- Data: ImageNet-1K; Pile subset (~1/20 of 300B tokens); PanGu π ~1.6T tokens.
- Eval: OpenCompass (HellaSwag, ARC-e, PIQA, C3, CSL, LAMBADA); PanGu π 's 11 benchmarks.
- Checkpointing: every epoch (ViT), every 1000 iters (LLMs); periodic resume.

Results

- Pythia-410M: ~70 $\times$ compression; Avg 42.93 vs 43.11 (-0.18) across HellaSwag/ARC-e/PIQA/C3/CSL/LAMBADA; size 4.53G $\rightarrow$ 0.06G.
- ViT-L/32: Top-1 $\approx$ 71.69% with both M2W+W2M; end-to-end CR (W+mom) up to ~35 $\times$; prior baselines ignore momentum.
- PanGu π -1B: 14.64G $\rightarrow$ 0.59G (~25 $\times$); Avg 40.83 vs 40.74 (+0.09).
- PanGu π -7B: 98.59G $\rightarrow$ 4.10G (~24 $\times$); Avg 57.09 vs 56.71 (+0.38).

Curves

- Loss curves closely match originals; small, stable gap.
- Checkpoint size tracks LR: larger at high LR, smaller at low LR (residual sparsity).

Figure 4: Loss over iters; per-checkpoint size vs iteration

Figure 4. Fidelity of training dynamics and per-node size reduction.

Analysis

Faithful recovery

Notes

- Compression runs independent of training; keep last reconstructed checkpoint.
- After training: keep init seed + compressed deltas to reconstruct any checkpoint.
- Hyperparameters: $\alpha = 5e-5$, $\beta = 2.0$, $n = 4$ (int4).

Usage

Deployment and recovery

Conclusions

- Extreme checkpoint compression with near-lossless performance.
- Residual encoding, joint weight–momentum pruning, non-uniform quantization.
- Up to $\sim 70\times$ reduction (Pythia-410M); strong on 1B–7B models.

Summary

Efficient and accurate

Future

- Extend to multimodal and vision LMs; include transformers/CNNs/RNNs.
- Refine theory with newer Adam analyses.
- Impact: improves ML efficiency; no specific concerns highlighted.

Outlook

Broader applicability

References

- Brown et al., 2020 (GPT-3); Touvron et al., 2023 (LLaMA); Chowdhery et al., 2023 (PaLM).
- Han et al., 2015 (DeepCompression); Dettmers et al., 2022; Xiao et al., 2023 (SmoothQuant).
- Chen et al., 2020 (LC-Checkpoint); Eisenman et al., 2022 (Check-N-Run); Hu et al., 2020 (Delta-DNN).
- Jin et al., 2023 (QD-Compressor); Agrawal et al., 2023 (DynaQuant); Kingma & Ba, 2014 (Adam).
- Wang et al., 2023 (PanGu π); Gao et al., 2020 (Pile); Dosovitskiy et al., 2020 (ViT).

Refs

See paper for full list

Code

Repository	https://github.com/Gaffey/ExCP
Conference	ICML 2024, PMLR 235
Contact	xinghao.chen@huawei.com; yunhe.wang@huawei.com

Links

Open-source release